Project 1: High-dimensional model-assisted estimation

Kalil Bouhadra, Aibat Kossumov, Jiayin Chen, Adrien da Silva

Abstract

This report investigates the behavior of model-assisted estimators for finite population means in high-dimensional settings where the number of covariates p is not negligible relative to the sample size n, i.e., $p/n \not\approx 0$. Using a simulation framework that allows p to grow with n, we compare the performance of several model-assisted estimators, each employing a different estimators in the first stage: linear regression, lasso, elastic net, random forest, gradient boosting. We also compare these estimators against the benchmark Horvitz-Thompson estimator to quantify efficiency gains. Our results illustrate how the choice of regression method influences the robustness of model-assisted estimation in high-dimensional contexts.

1 Introduction

In the absence of auxiliary information, the population total of a survey variable Y over a population $\mathcal{U}$ of size N is estimated using the Horvitz–Thompson estimator. Although design-unbiased, this estimator can have a large variance in high-dimensional setting. When auxiliary information is available, efficiency can be improved by incorporating it into the estimation procedure. In particular, exploiting the relationship between Y and the auxiliary variables $\mathbf{x}_k$ leads to more efficient estimators of the population total. In this context, the resulting estimators are referred to as model-assisted estimators.

Let p denote the number of auxiliary variables and n the sample size. The literature on model-assisted estimation has mainly focused on fixed number of covariates p. However, with the advent of big data, it is now common to handle datasets where the number of covariates is comparable to the sample size. In this report, we investigate the behavior of model-assisted estimators when p/n ranges from 0 to 2. Through a simulation study that considers both fixed-size (SRSWOR) and random-size (Bernoulli) sampling designs, we compare a range of regression methods (linear regression, lasso, elastic net) and machine learning methods (random forests and gradient boosting) for estimation in the first stage. Our goal is to assess how the choice of method and the sampling design influence the efficiency of model-assisted estimation in high-dimensional contexts, i.e., $p/n \not\approx 0$.

2 Setup

2.1 Finite Population Setup

Consider a finite population $\mathcal{U} = \{1, 2, \ldots, N\}$ of size N. For each unit $k \in U$, let y_k denotes the value of the survey variable for unit k in the population. We are interested in estimating the population mean

$$\mu_y = \frac{1}{N} \sum_{k \in \mathcal{U}} y_k.$$

To this end, we select a sample $\mathcal{S} \subset \mathcal{U}$ of size $n = |\mathcal{S}|$ according to a sampling design $P(\mathcal{S})$, which induces the first-order inclusion probabilities $\{\pi_k\}_{k \in \mathcal{U}}$ and the second-order inclusion probabilities $\{\pi_{k\ell}\}_{k,\ell \in \mathcal{U}}$.

2.2 Horvitz–Thompson and Model-Assisted Estimation

The classical design-based estimator of the population mean is the Horvitz–Thompson (HT) estimator

$$\hat{\mu}_{ht} = \frac{1}{N} \sum_{k \in \mathcal{S}} \frac{y_k}{\pi_k}.$$

It is the best linear design-unbiased estimator under the assumption that $\pi_k > 0$ for all $k \in \mathcal{U}$. However, it does not exploit the auxiliary variables $\mathbf{x_k}$ that may be available for each unit in the population. When the study variables y_k is strongly related to these covariates, ignoring this information may result in estimators with unnecessarily large variance. From now on, we assume that $\mathbf{x_k} = (x_{1k}, x_{2k}, \ldots, x_{pk})^\top$ is available $\forall k \in \mathcal{U}$. Thus, the observed data are

$$D_{ma} := \{(\mathbf{x_k}, y_k); k \in \mathcal{S}\} \cup \{\mathbf{x_k}; k \in \mathcal{U} \setminus \mathcal{S}\}$$

Let $m(\mathbf{x}) = E(y \mid \mathbf{x})$ be the true regression function and $\widehat{m}(\mathbf{x})$ be an estimator of it. Then, we define the model-assisted estimator based on $\widehat{m}$ to be

$$\hat{\mu}_{ma}(\widehat{m}) = \frac{1}{N} \sum_{k \in \mathcal{U}} \widehat{m}(\mathbf{x_k}) + \frac{1}{N} \sum_{k \in \mathcal{S}} \frac{y_k - \widehat{m}(\mathbf{x_k})}{\pi_k}.$$

The first term is a prediction component which estimates the population mean using model-based predictions. The second term adjusts for prediction errors using a design-weighted residual term. This structure ensures that the estimator remains approximately design-unbiased while potentially achieving substantial variance reduction when the prediction model is accurate.

3 Simulation

This section describes the Monte Carlo experiment used to compare the Horvitz–Thompson estimator and several model-assisted estimators of the finite population mean introduced in the previous section.

3.1 Simulation design

For each simulation scenario, we generate a finite population $\mathcal{U} = \{1, \ldots, N\}$ of size N with a study variable Y taking values $y_k \in \mathbb{R}$ and a collection of auxiliary variables represented by vectors $\mathbf{x}_k \in \mathbb{R}^p$ for each unit $k \in \mathcal{U}$. The purpose of the simulation is to assess how the performance of model-assisted estimators changes as the dimension of the auxiliary information increases. In all simulations, the population size N and the sample size n (or the expected sample size n_{exp}, in the case of random size sampling design) are considered to be fixed. For each scenario $p/n \in [0, 2]$, the true population is generated once and then kept fixed throughout all Monte Carlo repetitions. In the case of sampling designs with random size, scenarios are instead defined through $p/n_{\text{exp}} \in [0, 2]$. This makes it possible to isolate the effect of the sampling design and the estimation procedure from the randomness in the population generation step.

Population size and sample size. In the experiments reported here, the population size is fixed at $N = 500$. In the case of SRSWOR, the sample size is chosen as a fixed proportion of the population size, namely $n = 0.3N = 150$. In the case of Bernoulli sampling, the expected sample size is also set to $n_{\text{exp}} = 0.3N = 150$.

Relevant and noise covariates. For each unit $k \in \mathcal{U}$, the auxiliary vector $\mathbf{x}_k$ is partitioned into two components $\mathbf{x}_k = (\mathbf{x}_{1k}, \mathbf{x}_{2k})$, where:

- $\mathbf{x}_{1k} \in \mathbb{R}^{p_{\text{true}}}$ contains the truly relevant covariates that enter the data-generating mechanism of the response,
- $\mathbf{x}_{2k} \in \mathbb{R}^{p_{\text{noise}}}$ contains noise covariates that are observed but do not affect the response variable.

Hence, the total number of auxiliary variables is $p = p_{\text{true}} + p_{\text{noise}}$ and the corresponding dimension ratio is

$$\frac{p}{n} = \frac{p_{\text{true}} + p_{\text{noise}}}{n}.$$

To vary the ratio $p/n = p/150$ from 0 to 2, the numbers of relevant and noise covariates are chosen across simulation scenarios as follows:

$$(p_{\text{true}}, p_{\text{noise}}) \in \{(10,4), (21,9), (32,14), (42,18), \ldots, (200,86), (210,90)\}.$$

These choices ensure that, for each fixed value $a := p/n \in [0,2]$, the proportion of relevant covariates is 70% of p (i.e., $p_{\text{true}} = 0.7p$), while the remaining 30% correspond to noise covariates (i.e., $p_{\text{noise}} = 0.3p$). This framework allows us to investigate estimator performance as the problem moves from a low-dimensional regime to a moderately high-dimensional regime.

Generation of the auxiliary variables. The components of the relevant covariates are generated as independent standard Gaussian variables:

$$x_{1k,j} \overset{iid}{\sim} \mathcal{N}(0,1), \qquad j = 1, \ldots, p_{\text{true}}, \quad k = 1, \ldots, N.$$

The components of the noise covariates are generated independently according to a Student distribution with 3 degrees of freedom:

$$x_{2k,j} \overset{iid}{\sim} t_3, \qquad j = 1, \ldots, p_{\text{noise}}, \quad k = 1, \ldots, N.$$

The use of heavy-tailed noise variables introduces additional variability and makes the estimation problem more challenging, especially in higher-dimensional settings.

Generation of the study variable. The response variable $y_k \in \mathbb{R}$ is generated from a linear model involving only the relevant covariates:

$$y_k = \beta_0 + \sum_{j=1}^{p_{\text{true}}} \beta_j x_{1k,j} + \varepsilon_k, \qquad k = 1, \ldots, N,$$

where the regression coefficients are drawn once per scenario according to

$$\beta_j \overset{iid}{\sim} \text{Unif}(-1,1), \qquad j = 0, 1, \ldots, p_{\text{true}},$$

and the random errors satisfy

$$\varepsilon_k \overset{iid}{\sim} \mathcal{N}(0,1).$$

Therefore, only the variables in $\mathbf{x}_{1k}$ are informative for predicting y_k, whereas the variables in $\mathbf{x}_{2k}$ act as irrelevant covariates.

Sampling designs. For each generated population, two sampling designs are considered. First, under simple random sampling without replacement (SRSWOR), a sample $\mathcal{S}_{\text{SRSWOR}}$ of fixed size n is drawn uniformly from the population. In this case, the inclusion probabilities are $\pi_k = n/N = 150/500 = 0.3$ for all $k \in \mathcal{U}$, and the Horvitz–Thompson estimator reduces to the sample mean

$$\hat{\mu}_{HT}^{\text{SRSWOR}} = \frac{1}{n} \sum_{k \in \mathcal{S}_{\text{SRSWOR}}} y_k.$$

Second, under Bernoulli sampling, each unit $k \in \mathcal{U}$ is independently included in the sample with probability π. The resulting sample is

$$\mathcal{S}_{\text{Bern}} = \{k \in \mathcal{U} : I_k = 1\}, \qquad I_k \overset{\text{i.i.d}}{\sim} \text{Bernoulli}(\pi),$$

In this case, the sample size $|\mathcal{S}_{\text{Bern}}| = \sum_{k=1}^{N} I_k$ is random. Nevertheless, the expected sample size $n_{\text{exp}} = E\big(|\mathcal{S}_{\text{Bern}}|\big) = N\pi = 500 * 0.3 = 150$ is fixed. The corresponding Horvitz–Thompson estimator is

$$\hat{\mu}_{HT}^{\text{Bern}} = \frac{1}{N\pi} \sum_{k \in \mathcal{S}_{\text{Bern}}} y_k.$$

3.2 Estimation procedure

For each Monte Carlo repetition and for each sampling design, we compute the Horvitz–Thompson estimator and several model-assisted estimators. More precisely, the estimation procedure is as follows.

1. Generate a finite population $\mathcal{U} = \{1, \ldots, N\}$ together with the response values $(y_k)_{k \in \mathcal{U}}$ and the auxiliary vectors $(\mathbf{x}_k)_{k \in \mathcal{U}}$.
2. Draw a sample $\mathcal{S}$ according to one of the two sampling designs described above, and determine the corresponding inclusion probabilities π_k.
3. Compute the Horvitz–Thompson estimator of the population mean:
$$\hat{\mu}_{HT} = \frac{1}{N} \sum_{k \in \mathcal{S}} \frac{y_k}{\pi_k}.$$
4. Using only the sampled observations $\{(\mathbf{x}_k, y_k) : k \in \mathcal{S}\}$, fit a prediction model $\widehat{m}(\cdot)$ for the regression function $m(\mathbf{x}) = \mathbb{E}(y \mid \mathbf{x})$.
5. Use the fitted model to obtain predicted values $\widehat{m}(\mathbf{x}_k)$ for all units in the population.
6. Compute the model-assisted estimator
$$\hat{\mu}_{MA}(\widehat{m}) = \frac{1}{N} \sum_{k \in \mathcal{U}} \widehat{m}(\mathbf{x}_k) + \frac{1}{N} \sum_{k \in \mathcal{S}} \frac{y_k - \widehat{m}(\mathbf{x}_k)}{\pi_k}.$$

In practice, we consider the following prediction methods for constructing $\widehat{m}$:

- ordinary least squares (OLS),
- random forest (RF),

- gradient boosting machine (GBM),
- LASSO,
- Elastic Net.

Thus, for each sampling design, we compare one purely design-based estimator (HT) and five model-assisted estimators based on different predictive methods.

Implementation details. For OLS, the linear model includes an intercept term. When the number of covariates is too large relative to the sample size, OLS may become unstable or undefined; in such cases, the corresponding estimate may be unavailable. For LASSO and Elastic Net, the regularization parameter is selected by 5-fold cross-validation. For Elastic Net, the mixing parameter is fixed at $\alpha = 0.5$. For GBM, the model is fitted with Gaussian loss, 500 trees, interaction depth 3, shrinkage parameter 0.05, bag fraction 0.8, and minimum terminal node size equal to 3.

Monte Carlo repetitions For each scenario, the sampling and estimation steps described above are repeated independently $R = 200$ times. This yields, for each estimator and each design, a collection of Monte Carlo replicates

$$\hat{\mu}^{(1)}, \hat{\mu}^{(2)}, \ldots, \hat{\mu}^{(R)}.$$

3.3 Evaluation criteria

The main performance criterion is the mean squared error (MSE) with respect to the true finite population mean $\mu_y = \frac{1}{N}\sum_{k\in\mathcal{U}} y_k$. For an estimator $\hat{\mu}$, the empirical MSE over R Monte Carlo repetitions is defined as

$$\mathrm{MSE}(\hat{\mu}) = \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\mu}^{(r)} - \mu_y\right)^2 = \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\mu}^{(r)} - \overline{\mu}_y\right)^2 + (\overline{\mu}_y - \mu_y)^2,$$

where $\overline{\mu}_y = \frac{1}{R}\sum_{r=1}^{R}\widehat{\mu}^{(r)}$. This criterion captures both bias and variability, making it suitable for comparing the different estimators.

4 Results and Discussions

Figure 1 displays the MSE for the different regression methods tested under both Bernoulli sampling and SRSWOR. The results for the two sampling methods are very similar. However, some differences are present in the plots and will occasionally be highlighted in the following discussion.

We can immediately observe that as the ratio between the dimension and the sample size, p/n, approaches 1, the linear regression estimator begins to perform poorly. This is an expected result. As $p \to n$ from below, the inverse of the matrix $X^\top X$ becomes highly unstable. This behavior is more pronounced in the Bernoulli sampling case because the sample size is random.

Furthermore, we observe that the Horvitz–Thompson estimator has the highest MSE for all values of p/n tested, for both Bernoulli sampling and SRSWOR, compared to the other model-assisted estimators. This is again an expected result, since the Horvitz-Thompson

estimator does not incorporate any auxiliary information, leading to a larger MSE. Additionally, a model-assisted estimator can be viewed as equivalent to the Horvitz–Thompson estimator when the model provides no useful information, and as an improvement over it when the chosen model captures some underlying structure of the data.

We also observe that the estimator based on LASSO and Elastic Net regression has the lowest MSE. In our simulation, the hyperparameter for the LASSO regression model was $\alpha = 1$. This hyperparameter controls L_1 regularization. Since LASSO prefers a sparse model, we would expect it to ignore the noise covariates, and have lower MSE, which is what we observe in Figure 1. Elastic Net combines both LASSO regression (L_1 regularization) and Ridge regression (L_2 regularization). We observe that Elastic Net performs similarly to LASSO, which is what we would expect.

Finally, Figure 1 suggests that the estimators based on Random Forest and GBM exhibit larger MSE than those based on LASSO or Elastic Net. A possible explanation is that these methods tend to overfit the noise covariates, whereas LASSO and Elastic Net are more likely to shrink the corresponding coefficients to zero. This observation is closely related to the data-generating process, which follows a linear model structure with additional noise covariates. This feature directly impacts the performance of the model-assisted estimators when predicting from sample observations. In particular, it explains why penalized variants of linear model, achieve lower MSE compared to more flexible methods such as Random Forest/GBM, as they are better specified for this setting.

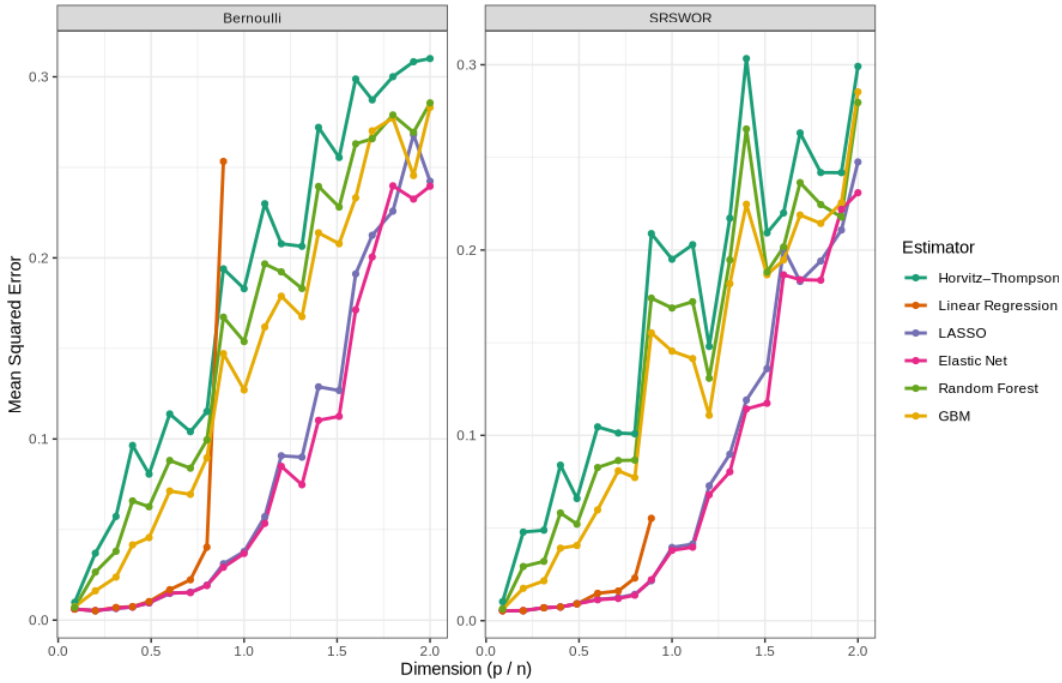

Figure 1: Mean Squared Error of different estimators as a function of the dimension-to-sample-size ratio (p/n), under Bernoulli sampling (left) and SRSWOR (right). The Horvitz–Thompson estimator exhibits the highest MSE, while penalized linear models (LASSO and Elastic Net) generally achieve better performance than more flexible methods such as Random Forest and GBM.

Concerning the decomposition of the MSE into bias and variance, we observe that the bias is negligible. Therefore, the plots of the MSE are very similar to those of the variance.

In Figure 2, we present the boxplots obtained from the Monte Carlo simulations for a specific case where $p/n = 0.4$. These simulations allow us to assess the sampling variability of the estimators across repeated samples. First, we observe that for each estimator, the empirical mean and median are close to each other, suggesting that the underlying distributions are approximately symmetric.

Furthermore, the empirical means of all estimators are very close to the true population mean, indicating that these estimators exhibit negligible empirical bias in this specific scenario where $p/n = 0.4$. Also note that the boxplots corresponding to the linear regression model and its penalized variants exhibit substantially lower variability than other estimators. Their distributions are more concentrated, with tighter quantiles compared to the other methods.

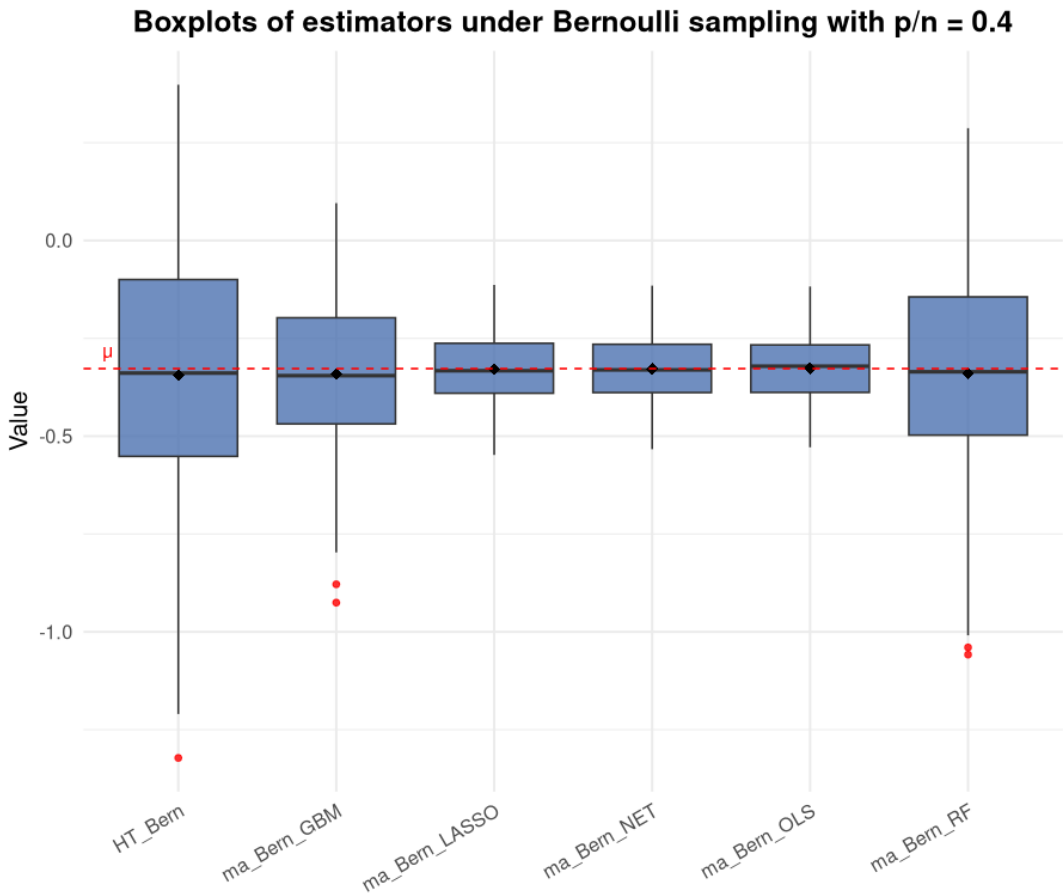

Figure 2: Boxplots of the estimators obtained from Monte Carlo simulations under Bernoulli sampling with $p/n = 0.4$. The red horizontal line indicates the true population mean. The black horizontal line inside each boxplot represents the sample median of the corresponding estimators, while the black dot represents the sample mean.

5 Conclusion

In this report, we studied the behavior of model-assisted estimators in high-dimensional settings where the ratio p/n is not negligible. Overall, these estimators maintain good properties, with very small empirical bias even when p is comparable to n. However, their

performance in terms of MSE strongly depends on how well the prediction model captures the relationship between Y and the auxiliary variables. When the model is informative, clear efficiency gains are observed compared to the Horvitz–Thompson estimator. Across all scenarios, the bias remains negligible compared to the variance for all estimators.

As the dimension increases, the difficulty of the problem becomes more apparent. In particular, ordinary least squares becomes unstable when p approaches n, leading to high variability. This effect is even more noticeable under Bernoulli sampling due to the randomness in the sample size. More generally, a larger number of covariates increases the influence of noise variables, which can negatively impact prediction and estimation accuracy.

Some methods looks more robust to high dimensionality than others. Penalized approaches such as LASSO and Elastic Net perform well when p/n is large. By shrinking coefficients and selecting relevant variables, they limit the impact of noise and remain stable. In contrast, more flexible methods like Random Forest and Gradient Boosting tend to be more affected by noisy covariates in this setting, which leads to higher MSE in our simulations.

The work completed in this project shows the importance of model-assisted estimator when working with high-dimensional data, as all models outperformed the standard Horvitz-Thompson estimator. Further work in this area could include testing other regression functions, and could include increasing the dimension ratio p/n since we only tested p/n from 0 to 2.